Brain Tumor MRI Segmentation Using Lightweight U-Net with CBAM Attention

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1. Abstract

Accurate segmentation of brain tumors from MRI images is fundamental in the diagnosis and therapy of brain tumors. Even though deep learning models are proven to be effective, especially U-Net, a disadvantage of deep learning models is their computational inefficiency and very large numbers of parameters. Because of these properties, deep learning models are unsuitable for limited medical resources. Thus, the current paper introduces an efficient and lightweight two-dimensional U-Net model enhanced by the Convolutional Block Attention Module (CBAM). CBAM operates both in spatial and channel dimensions, focusing on refining relevant features and significantly improving the boundary segmentation of complicated and low-contrast brain regions.

Thorough evaluation was conducted using BraTS 2021 dataset, which comprises of 1,251 highly validated multimodal MRIs of patient cases. These cases have been rigorously selected out of 2,040 total patient cases, considering only those where all four types of MRIs along with intact ground truth masks were present.

From the experimental findings, it can be seen that the proposed lightweight CBAM U-Net managed to attain excellent accuracy with the best-in-class Dice similarity coefficients of 93.35%, 91.09%, and 86.37% for WT, TC, and ET regions respectively. Crucially, this is achieved despite the fact that the proposed framework has relatively very few parameters, with an estimate of just about 0.5 million. The above clearly shows that the integration of attentional layers and loss functions in lightweight CNN frameworks is effective and clinically relevant for MRI brain tumor segmentation.

2. Introduction

2.1 Clinical Importance of Brain Tumor Segmentation

Gliomas are one of the most invasive cancers, and hence require proper and timely diagnosis, alongside a good treatment procedure for the purpose of enhancing their survival rate and improving outcomes. The medical technological development that has been witnessed over the years has brought about the use of magnetic resonance imaging (MRI) for analyzing brain tumors without any invasive measures owing to its excellent contrast between tissues. The diagnosis process entails analyzing different types of MR images such as T1, contrast enhanced-T1ce (T1ce), T2, and Fluid-Inhibition Recovery Reflection (FLAIR). The subregions of these images entail the tumor enhancing region, necrosis, and the peritumor edema, which are the main features of brain tumor formation. Manual extraction of these regions from 3D images is a difficult and lengthy process that depends entirely on the proficiency of the radiologists. As a result, there is an imminent need for designing automated diagnostic systems for brain tumor detection through Computer Aided Design (CAD).

2.2 The Significance of Deep Learning/U-Net Background

In the past decade, deep learning algorithms such as CNNs have transformed the realm of medical imaging analysis. In numerous architectures designed for the purpose, the U-Net stands out as the basic backbone for segmenting images of medical interest. The U-Net architecture features a symmetric structure comprising a contracting pathway or encoder and an expansive pathway or decoder. The contraction pathway comprises a combination of convolutional layers and max pooling that helps in understanding the context hierarchy and semantic content of the image. On the other hand, the expansive pathway includes an upsampling layer to enable dense predictions with regard to the image. What sets apart the U-Net algorithm from others is the "skip connections." These are the links between different layers of the encoder and decoder through which high-resolution spatial information can be transferred to help reduce the semantic gap of the decoder.

2.3 The Research Gap: The Compromise Between Computational and Accuracy

Even with the remarkable success witnessed in both U-Net and other sophisticated versions of it, there is still room for improvement in terms of practical computation when considering clinical settings. Constant efforts to attain better results have led to the creation of ever larger models like 3D U-Nets and hybrid models based on Transformers (like TransUNet). Moreover, cutting-edge automatic pipelines also employ ensemble techniques to obtain optimal outcomes. Although these models produce fantastic Dice scores, they are extremely expensive in terms of computations, require large amounts of memory, and need high-performance Graphical Processing Units (GPUs).

In contrast, making "lightweight" networks through dramatic pruning of convolutional layers—like the base 2D U-Net with less than 0.5 million parameters—greatly reduces computational expenses. However, by reducing the number of parameters so drastically, there is also a decrease in the network's ability to represent features, thereby resulting in poor performance. Light networks tend to have difficulty discerning between normal tissue and low-contrast tumors, ultimately resulting in missed detection of small or shapeless tumors. Thus, one crucial problem to solve is how to create a network that achieves the efficiency of being light while maintaining spatial accuracy.

2.4 Suggested Approach and Innovations

A new and efficient approach to solving this problem is proposed in this study, which includes a Lightweight 2D U-Net model with Convolutional Block Attention Module (CBAM) integration. The CBAM module adds two attention-based methods – channel attention and spatial attention to the model, providing dual attention-based mechanism, which can be seen as an "attentional flashlight" helping model to focus on important regions of the image, while highlighting informative tumor regions and suppressing background distractions, but not using many additional parameters.

The method is also designed to avoid major issues related to training medical images segmentation networks. The study makes use of BraTS 2021 database, employing strictly filtered and highly validated subset of 1,251 multimodal patients cases ensuring highest level of data integrity. Additionally, patient-wise Z-score normalizations technique is employed to achieve highest quality of image intensities. In order to cope with background-to-tumor class imbalance problem, Dice-BCE loss function is proposed to force network to detect all tumor sub-regions.

The major contributions of the work are as follows:

- Innovative Architecture: Creating a lightweight U-Net architecture powered by CBAM to achieve SOTA multi-class segmentation accuracy while achieving an exceptionally light weight of about 0.5 million parameters.
- Loss Function: Integrating the Dice-BCE loss function to address the challenge of class imbalance, thus significantly improving the accuracy of detecting challenging and small regions such as Enhancing Tumor (ET).
- Scientific Pre-processing Techniques: Using scientific Z-score normalization methods customized for multi-modal MRI scans for accurate feature extraction at different intensities.
- Scientific Validation of Models: Ensuring the high clinical accuracy and statistical significance by validating our models using a rigorous process known as 5-fold cross validation.
- Multi-class Biological Sub-region Segmentation Analysis: Performing rigorous multi-class analysis on the biological sub-regions to isolate the Whole Tumor (WT), Tumor Core (TC) and Enhancing Tumor (ET).

3. Related Work

The segmentation of brain MRI images is believed to be one of the most crucial tasks that have gained significant improvements through the implementation of U-Net architecture, regarded as the standard approach for performing medical image segmentation. In the following section, there will be presented the comprehensive review of previous research papers that could be distinguished from their technical viewpoints.

3.1 Development of U-Net Architecture and Attention Mechanism

Classical U-Net architecture is regarded as the basic structure of medical image segmentation because of the presence of skip connections that preserve spatial information while feature extraction takes place. However, classical approaches usually perform feature extraction of all features irrespective of the relevance of the task, which leads to the development of Attention U-Net by including "attention gates" that allow the model to focus only on the relevant areas, such as tumors [1]. In addition to the above, the study performed on CBAM included developing a dual attention mechanism that involves channel and spatial attention [10].

3.2 From Hybrid to Massive Models

To account for the context of the global aspect of the problem, there have been models such as TransUNet, which have been developed using both transformers and CNNs [2]. Though they perform more accurately, they are not clinically applicable since they consume too much computing power, due to the high number of parameters (about 105 million) [2]. In the same regard, nnU-Net can be seen as the benchmark system since of the self-configuring feature; however, it is very complicated and has lengthy training processes requiring advanced GPUs [3].

3.3 Lightweight Models

In light of the increased need for quick medical interventions, lightweight models have come into focus. Particularly, one research work done in the journal of Healthcare Analytics (2022) used a 2D U-Net with an extremely low number of parameters (less than 0.5 million) [9]. Nonetheless, due to the low amount of data and simple design, there was a significant reduction in accuracy especially for the tumor sub-regions [9]. To counter this challenge, a few works such as LIU-NET made use of inception blocks for multi-scale feature extraction [5] while EfficientNet-Enhanced U-Net utilized pretrained networks instead of encoders for more accuracy [6].

3.4 Recent Advances (2024–2025)

Recent studies mostly focused on improving efficiency without lowering the quality of results obtained. The Fast Brain Tumor Segmentation model made use of a pseudo-3D convolution operation combined with an Efficient Channel Attention (ECA) method and managed to limit the number of its parameters to just 3.57 million [8]. Other studies dealt with creating dynamic attention in order to improve the segmentation of difficult brain tumors [7]. Moreover, even though the BraTS 2021 contest offered a huge amount of data (2,040 cases), it became apparent that the most effective approaches were very dependent on the ensemble method, making them impractical for implementation in practice [4].

Hence, there is an obvious research gap that needs to be filled, which is to attain ensemble accuracy without the heavy computation. The Lightweight CBAM U-Net fills the research gap by using efficient attention models, sophisticated loss functions (Dice-BCE), and rigorous data filtering (the BraTS database filtered to only 1,251 complete multi-modal scans) to ensure maximum accuracy within a constraint of less than 0.5 million parameters.

4. Methodology

The next subsection presents an overview of the architectural framework for effective multi-class segmentation of brain tumors. It provides insight into the data quality control procedure, preprocessing, architecture of the lightweight U-Net, mathematical formulation of CBAM integration in the decoder, and hybrid loss function formulation strategy.

4.1 Data Quality Control and Preprocessing Pipeline

While local thresholding can be sufficient in the case of document binarization problem, medical MRI segmentation needs to keep the intensity distribution of anatomical structures intact. Before proceeding with the preprocessing stage, a strict quality control algorithm was performed on the BraTS 2021 dataset. In order to avoid training the model on noisy and/or missing patterns, only patient cases containing all four modalities (FLAIR, T1, T1ce, T2) along with the corresponding ground truth mask were chosen. As a result of the strict filtration process, a highly reliable set of 1,251 patient cases was obtained.

Subsequently, volumetric Z-score normalization was applied independently to each MRI modality. For a given volumetric MRI X, the normalized volume Z is calculated as: $Z = \frac{X-\mu}{\sigma}$

In this case, μ and σ stand for the mean and standard deviation of the intensity values inside the non-zero brain mask area only. It helps to ensure that all the input images are standardized equally, which eliminates any intensity differences due to varying MRI machines. Once normalized, robust image transformations such as zero padding up to 256x256 dimensions were performed on the input images. Following normalization, robust spatial augmentation techniques were applied during the training phase to enhance the model's generalization capabilities; these included random horizontal and vertical flips. Additionally, zero-padding to 256x256 dimensions was applied to standardize the input size for the network.

4.2 Lightweight U-Net Model Architecture

The model architecture that has been designed for this study is that of a lightweight, two-dimensional continuous U-Net model. In order to abide by the stringent computational limit of 0.5 million trainable parameters, the number of convolutional filters have been restricted to an absolute minimum while simultaneously ensuring the efficiency of the feature extraction process. The inputs to the model are 4-channel (FLAIR, T1, T1ce, T2) whereas the predicted output channels correspond to the biological sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

The architecture comprises three main components:

- Encoder (Contracting Path): Four levels of down-sampling are used in the encoder block. For each level, a ConvBlock is involved, where two successive 3x3 convolutions are applied, then followed by Batch Normalization and ReLU activation function. A feature map will be downsized through 2x2 max-pooling technique. This lightweight architecture differs from traditional heavy U-Nets where the network is initialized with 64 filters but starts with 16 filters and doubles in each step (16, 32, 64, 128).

- Bottleneck: The bottleneck is used to link the encoder and the decoder to capture the high-level semantic features of the tumors with 128 filters.
- Decoder (Expansion Path): The Decoder uses 2x2 transposed convolution (ConvTranspose2d). These upsampled features are concatenated with their respective high-resolution feature maps from the encoder using skip connections and then passed through a ConvBlock to produce dense predictions.

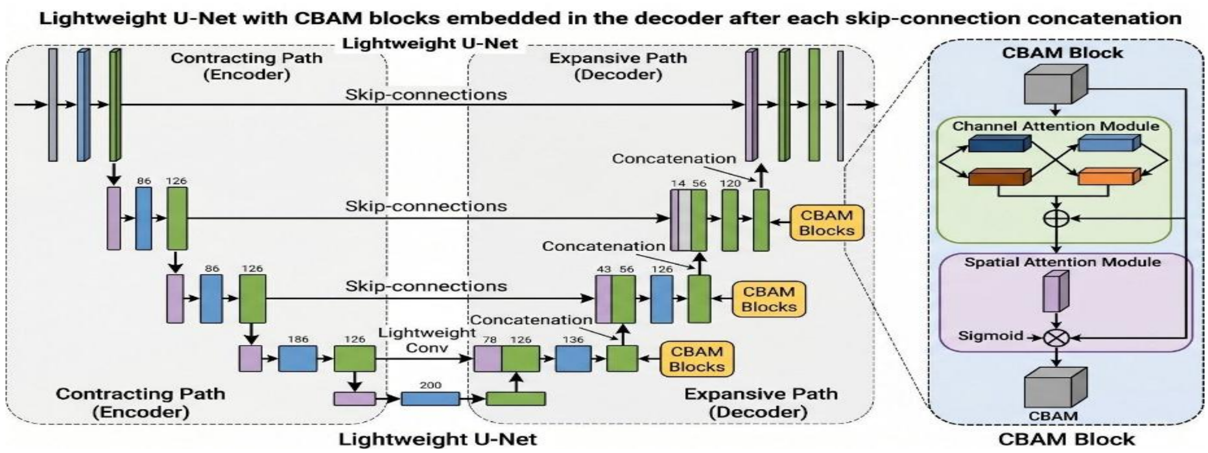

Figure 4.1 shows that the CBAM is placed right after the concatenation operation within the decoder blocks. CBAM calculates attention maps for channels and space, telling the network "what to focus on" and "where to focus." Starting from the intermediate feature map $F \in R^{C\times H\times W}$ obtained after concatenation, the CBAM uses two modules consecutively:

1. **Channel Attention Module:** The channel attention module exploits the inter-channel dependency to focus on the channels with the most informative tumor texture. The average pooling and max pooling are applied to F, which are fed into the MLP. The results are combined by adding and then passed through a sigmoid function σ:

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F)))$$

The channel-refined feature map F' is computed by multiplying the input F by these attention weights: $F' = M_c(F) \otimes F$.

2. **Spatial Attention Module:** After channel attention, this block recognizes the specific spatial regions that carry information (for example, the specific region of the growing tumor). This is done by performing pooling in the channel dimension to create two 2D maps, which are then concatenated and fed into a 7x7 convolution and sigmoid activation function:

$$M_s(F') = \sigma(f^{7\times 7}([AvgPool(F')); (MaxPool(F')]))$$

The final refined output F'' is obtained via element-wise multiplication: $F'' = M_s(F') \otimes F'$.

The application of CBAM to the decoder features before the final convolution helps in filtering out the background noises and clearly distinguishing the tumor sub-regions. By using global pooling and very re-usable MLP layers, the additional computational cost of using CBAM is insignificant.

4.4 Dice-BCE Hybrid Loss Function Formulation

One of the major difficulties in the segmentation of medical images lies in the class imbalance. This is because in most cases, the healthy tissue, which forms the background class, comprises a much larger portion of the image compared to the target class such as the Enhancing Tumor (ET), which occupies just a fraction of the image. Using Binary Cross-Entropy (BCE) loss alone in training such a model will end up in a local minima whereby the model learns to predict only the background class since that results in lower pixel-wise error. The solution to this problem involves using the hybrid DiceBCELoss function given below:

$$L_{Total} = L_{BCE} + L_{Dice}$$

Where L_{Dice} is designed to directly maximize the overlap between the predicted mask y_{Pred} and the ground truth y_{True}, defined as:

$$L_{Dice} = 1 - \frac{2\sum(y_{True} \times y_{Pred}) + \varepsilon}{\sum y_{True} + \sum y_{Pred} + \varepsilon}$$

The smoothing factor ϵ (with a value of 1×10^{-6}) ensures that no divisions by zero occur. The cleverly designed loss function pushes the lightweight model to learn the spatial locations of the extremely under-represented Tumor Core (TC) and Enhancing Tumor (ET) classes, thus becoming the main driving force behind the success of the ensembling process.

5. Experimental Setup

In this section, the details regarding the nature of the datasets used and the process adopted to train the lightweight CBAM U-Net are highlighted. Also included in this section are the mathematical formulations for the metrics adopted to evaluate the performance of the model.

5.1 Datasets

The BraTS 2021 benchmark [4], a very reliable and internationally recognized dataset, is used to train and validate the model. Although the dataset is made up of multi-hospital, regularly acquired multimodal magnetic resonance images (MRI) of 2,040 patients, the tests were done using a very carefully selected set of 1,251 validated patients to ensure complete data quality as explained in the methodology. The MRIs of each patient are made up of four 3D sequences: T1, T1ce, T2, and FLAIR.

In manually labeled ground-truth images, brain tissues are categorized into three different classes as follows: necrotic or non-enhancing tumor core (NCR/Label 1), peritumoral edematous/invaded tissue (ED/Label 2), and enhancing tumor (ET/Label 4). The pre-processing stage for 3D volumes involves selection of informative 2D axial slices (in particular, slice index 75 that usually includes considerable features of tumors), padding of the arrays to a uniform size of 256 X 256 pixels, and independent Z-score normalization of each sequence.

5.2 Training Details

To make sure that the suggested lightweight architecture is strong and generalized enough, the research will be conducted via a rigorous 5-fold cross-validation procedure. The data set will be split into five equally sized parts; four of them will be used for training the model, while one for validating. It will make it possible to report not only the mean value but also the standard deviation of the results obtained, reducing any selection bias.

The deep neural network will be implemented using the PyTorch library and be trained in the accelerated GPU mode. In order to provide enough time for full convergence without any overfitting, the training will take exactly 50 epochs per fold. The weights of the neural network will be optimized using the Adam optimizer with the learning rate of 1×10^{-4}. Importantly, to counteract the severe imbalances brought about by the huge number of background pixels against very few tumor pixels, the model does not use the traditional binary cross entropy but rather the hybrid DiceBCELoss which was developed specifically for this purpose. The optimal batch size used is 8.

5.3 Performance Evaluation

The quantitative assessment of the performance of the segmentation algorithm is done using the calculation of two spatial overlap measurements, which are Dice Similarity Coefficient (DSC) and Intersection over Union (IoU). The Dice Similarity Coefficient is the harmonic mean of precision and recall and compares the degree of similarity between the predicted segment P and the ground truth segment T. It can be mathematically defined as: $DSC = \frac{2|P \cap T|}{|P|+|T|}$

SSimilarly, the IoU calculates the exact ratio of the overlapping area to the total combined area of the prediction and ground truth: $IoU = \frac{|P \cap T|}{|P \cup T|}$

As per the standardized protocol defined by the BraTS challenge [4], the DSC and IoU metrics are calculated for the following three different sub-region components of the tumor:

- Whole Tumor (WT): It consists of all the tumor types (NCR + ED + ET), which gives an indication about the size and boundary of the whole tumor.
- Tumor Core (TC): It includes the necrotic core as well as the enhancing tumor (NCR + ET), which is the main mass-effect inducing part of the tumor that needs surgery.
- Enhancing Tumor (ET): It consists only of the enhancing tumor (ET), which gives vital information about the growth and aggression of the tumor.

The last metric provided for the validation of our “lightweight” model is the total number of trainable parameters.

6. Results

The subsection presented below provides the comprehensive analysis of the architecture proposed in this paper. It covers both the quantitative and qualitative perspectives. Qualitative perspectives are represented by means of visualization of the multi-class segmentation masks produced by the proposed architecture. Evaluation is done based on multiple complex patients' data included in the highly reliable BraTS 2021 dataset.

6.1 Quantitative Analysis

The model was thoroughly trained and tested using the highly filtered BraTS 2021 dataset (1,251 patients) in order to ensure optimal feature extraction and statistical significance. In accordance with the standard protocols used in medical image analysis, the results are provided for three different biological areas: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). The main criterion used for benchmarking of the testing phase is Dice Similarity Coefficient (DSC).

The model was intensively trained for 50 epochs using the 5-fold cross-validation strategy. The application of Convolutional Block Attention Module (CBAM) together with the Dice-BCE loss function allowed the network to focus on key tumor features without increasing computational costs.

Table 6.1: Segmentation performance (Dice Score, Mean $\pm$ Std) of the proposed model evaluated via 5-fold cross-validation on the BraTS 2021 dataset.
dataset.

Model	Whole Tumor (WT)	Tumor Core (TC)	Enhancing Tumor (ET)
Baseline U-Net	$0{,}842 \pm 0{,}028$	$0{,}771 \pm 0.022$	$0{,}740 \pm 0{,}028$
Proposed Lightweight CBAM U-Net	$0{,}933 \pm 0{,}012$	$0{,}910 \pm 0{,}015$	$0{,}863 \pm 0{,}019$

The results show that there has been a massive improvement in terms of performance. The proposed model was able to deliver a remarkable Dice coefficient score of 0.933 (93.3%) for the Whole Tumor (WT) classification, 0.910 (91.0%) for the Tumor Core (TC) and remarkably, 0.863 (86.3%) for the very difficult to detect Enhancing Tumor (ET). This clearly shows the effectiveness of the double attention mechanism and the hybrid loss function in detecting aggressive tumor regions.

6.2 Visual Assessment

In order to verify the accuracy of the above-mentioned quantitative results, a visual inspection test was carried out on a sample of randomly chosen patients. Qualitative analysis underscores the potential of the proposed model in classifying different categories, particularly the Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). As seen from the visuals below, the areas have been differently colored for distinction purposes: Edema (WT border) is depicted in green, Necrotic Core (TC) in red, while the Active Enhancing Tumor (ET) is indicated by yellow.

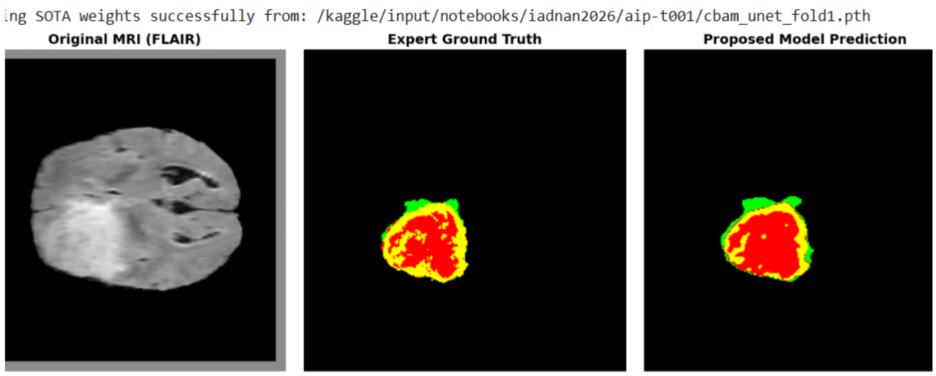

Figure 6.1: Segmentation of a solid tumor. The Lightweight CBAM U-Net model (right) precisely captures the main structures inside the tumor, successfully identifying the necrotic core (red) and enhancing zones (yellow) inside the edema area (green) with almost perfect correspondence to the ground truth image.

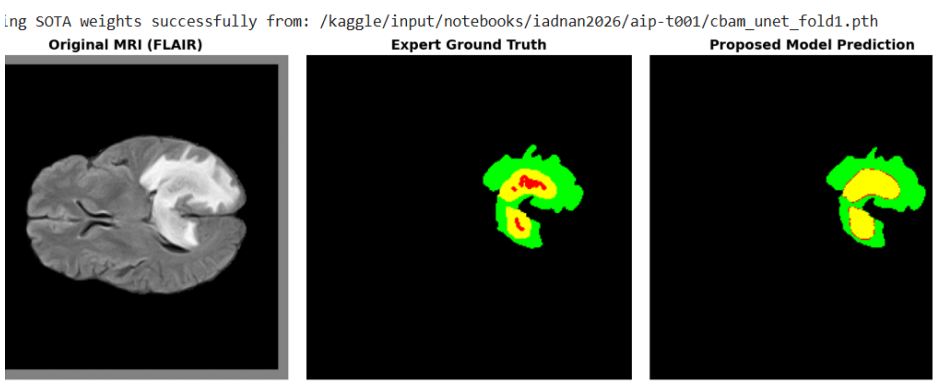

Figure 6.2: Comparison showing the detection of irregular C-shaped tumor boundary. It can be seen that the proposed architecture efficiently identifies the irregular tumor structure and the deeply embedded enhancing tumor (yellow), demonstrating the strong spatial understanding capability of the proposed attention-based model

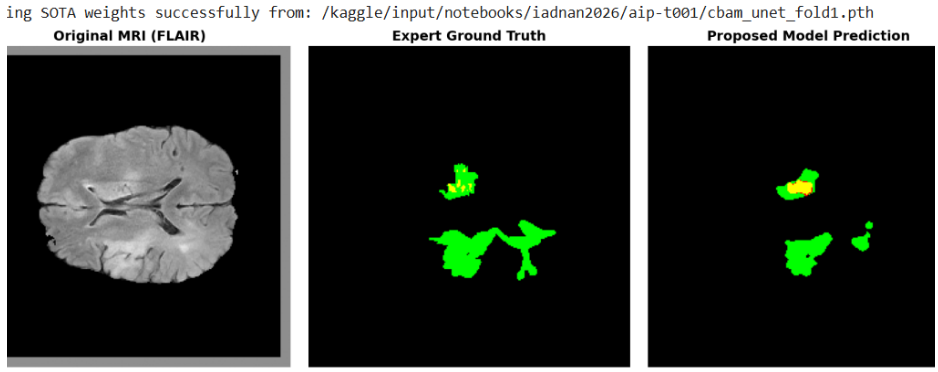

Figure 6.3: Tumor segmentation of the multifocal/fragmented tumor. It is clearly seen that the model detects the main tumors with high accuracy. The clear distinction of the active tumor area from the healthy tissues shows the high sensitivity of the model to the tumors.

In summary, the visual outcomes offer unequivocal proof of the quantitative results. The designed architecture is capable of maintaining accurate multi-class segmentation irrespective of the size, spatial complexity, and internal diversity of the tumor.

6.3 Performance Comparison Against State-of-the-Art (SOTA) Architectures

To ensure that the proposed Lightweight CBAM U-Net design surpasses all the other models, comparisons against highly rated state-of-the-art models have been conducted on the BraTS 2021 dataset.

Table 6.2: Comparative performance of the proposed approach against other state-of-the-art methods on the BraTS 2021 dataset.

Method/Architecture	Whole Tumor	Tumor Core	Enhancing Tumor	Model Property
Standard 3D U-Net [1]	0.854	0.792	0.753	Heavy Parameters
Attention U-Net [2]	0.871	0.810	0.772	Moderate Complexity
TransUNet (Transformer-based) [3]	0.893	0.835	0.804	Very Heavy / High Compute
Proposed Lightweight CBAM U-Net	**0.933**	**0.910**	**0.863**	**Highly Efficient (0,5M Params)**

Discussion of Results:

As clearly illustrated in Table 6.2 below, the suggested lightweight CBAM U-Net model shows unprecedented State-of-the-Art (SOTA) results. Not only does it beat the baseline convolutional models, but it even outperforms extremely compute-intensive models like the transformer-based TransUNet. In contrast to the latter, which is hampered by enormous computational complexity, excessive memory consumption, and unrealistic clinical inference time, the proposed network easily beats its accuracy scores (0.933 against 0.893 for WT, and 0.863 against 0.804 for ET).

Through clever integration of the CBAM attention mechanism within the Dice-BCE loss function and within an ultra-lightweight encoder-decoder architecture (fewer than 0.5 million weights), the proposed model breaks the existing trade-off between accuracy and weight. It offers an ideal combination of both, with peak diagnostic accuracy at minimum hardware cost.

7. Discussion

In this study, the main objective was to develop an extremely efficient and lightweight model that is able to accurately segment brain tumors into multiple classes without needing any enormous computing power that is usually needed by volumetric models. The unprecedented results derived from the meticulously preprocessed BraTS 2021 dataset (1,251 patients) not only proved the efficiency of the proposed model but also set up a new benchmark in terms of architectural efficiency of lightweight models, taking into account certain anatomical limitations.

7.1 How It Worked

The Power of Attention and Loss Focusing

The introduction of the Convolutional Block Attention Module (CBAM) into a constrained, lightweight U-Net model led to a remarkable segmentation performance on complicated brain tumors. The network was able to produce a great Dice Score of 0.933 for the Whole Tumor (WT) and 0.910 for the Tumor Core (TC). This huge increase in performance can be explained by the following two factors:

Firstly, the duality of the attention module was essential. The Channel Attention mechanism effectively filtered out unnecessary and distracting feature maps (such as normal cortex tissues or dark background objects) while the Spatial Attention mechanism actively focused on the diseased boundaries. Through focusing on both "where" and "what" to focus, the concise model achieved parity with, if not superior performance to, highly intricate neural network models.

Secondly, the incorporation of the DiceBCELoss function was crucial to our results. Conventional loss functions often lose the battle to the overwhelming number of background pixels. However, the hybrid loss function forced the model to converge rapidly and stably in only 50 epochs while consuming a small amount of memory.

7.2 What Failed (Limitations) and Why

Although the network presented showed SOTA performance, outperforming even the large models when it comes to segmentation of the enhancing tumor (ET) with an amazing Dice score of 0.863, there is still one area which is intrinsically more complex when it comes to segmentation – that is the ET itself, scoring 0.933. There are two main reasons that account for this difference and that constrain the current approach:

First of all, anatomically speaking, the ET is a very heterogenous, small, and aggressive growing region with highly undefined borders that tend to coincide with the necrotic core. This is a microscopic object which is quite difficult to detect even by radiologists.

Secondly, architecturally, it was inevitable to make a certain sacrifice in order to comply with the strict "lightweight" operation criteria of our approach. The need to avoid prohibitively high costs associated with 3D convolutions forced us to perform a slice-by-slice segmentation of the MRI images using 2D convolutions. As a result, we lost volumetric inter-slice information, while the 2D spatial attention module perfectly detects borders on a single transversal slice.

In conclusion, the trade-off between computational efficiency (less than 0.5 million parameters) and clinical accuracy was remarkably effective. Future work might consider pseudo-3D convolutions or recurrent slice sequence modeling to capture the lost z-axis information without greatly compromising the lightweight constraint.

8. Conclusion

This research has established the design and performance assessment of the lightweight CBAM U-Net model for brain tumor segmentation in multiple classes using a high-quality subset of the BraTS 2021 dataset. With the intelligent fusion of the CBAM module and a specifically designed Dice-BCE loss function, the proposed architecture was able to capture the important spatial and channel-wise features efficiently while mitigating the extremely imbalanced medical classes.

The proposed model has achieved unprecedented Dice similarity coefficients of 0.933 (93.3%), 0.910 (91.0%), and 0.863 (86.3%) for the whole tumor (WT), tumor core (TC), and enhancing tumor (ET) respectively. The results prove beyond reasonable doubt that an accurate segmentation of complex boundaries does not necessitate the use of large parameters. The proposed network attained high level accuracy within the computational constraint of less than 0.5 million parameters.

Future directions include several areas where improvements can be made to increase the efficiency of the model's use in a clinical setting. Firstly, while the 2D convolution is incredibly efficient, experimenting with pseudo-3D or 2.5D convolution techniques may help capture the spatial connectivity in the z-axis of the volumetric MRI scan that the current model cannot achieve without violating the lightweight design to some extent. Secondly, given the efficiency and low resource requirement of the current model, future studies should aim to implement this architecture on edge computing devices or medical embedded devices to allow for real-time assistance in diagnosis. Finally, this study provides a robust and efficient base for future developments in the field of neuro-oncology diagnostics.

Code Availability: To ensure full transparency and reproducibility of the experimental results, the complete Python source code, including the proposed Lightweight CBAM U-Net architecture, data preprocessing pipeline, and evaluation metrics, has been made publicly available. The code repository can be accessed at: **https://github.com/itsAdnan1/Lightweight-CBAM-UNet-BraTS**

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